

Q1

Consider a uniaxial single lamina of HS-carbon/epoxy, with a 60 vol.% fibre content.

Assuming that the Young's modulus of HS carbon and epoxy are respectively 450 GPa and 3.5 GPa:

Calculate the Young's Modulus of the composite along the fibre axis (E_{C1}).

$$E_{C1} = E_f V_f + E_m (1 - V_f) = 271.4 \text{ GPa}$$

Q2

Consider a uniaxial single lamina of HS-carbon/epoxy, with a 60 vol.% fibre content.

Assuming that the Young's modulus of HS carbon and epoxy are respectively 450 GPa and 3.5 GPa:

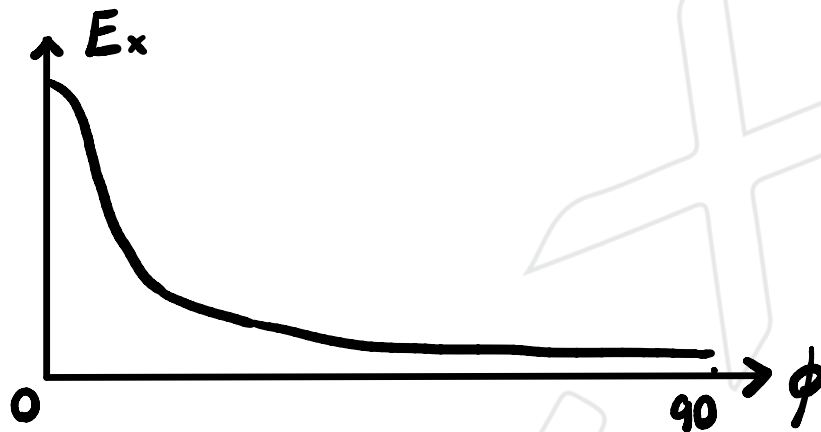
Calculate the Young's Modulus of the composite transverse to the fibre axis (E_{C2}).

$$\frac{1}{E_{C2}} = \frac{V_f}{E_f} + \frac{1-V_f}{E_m}$$

$$\therefore E_{C2} = 8.65 \text{ GPa}$$

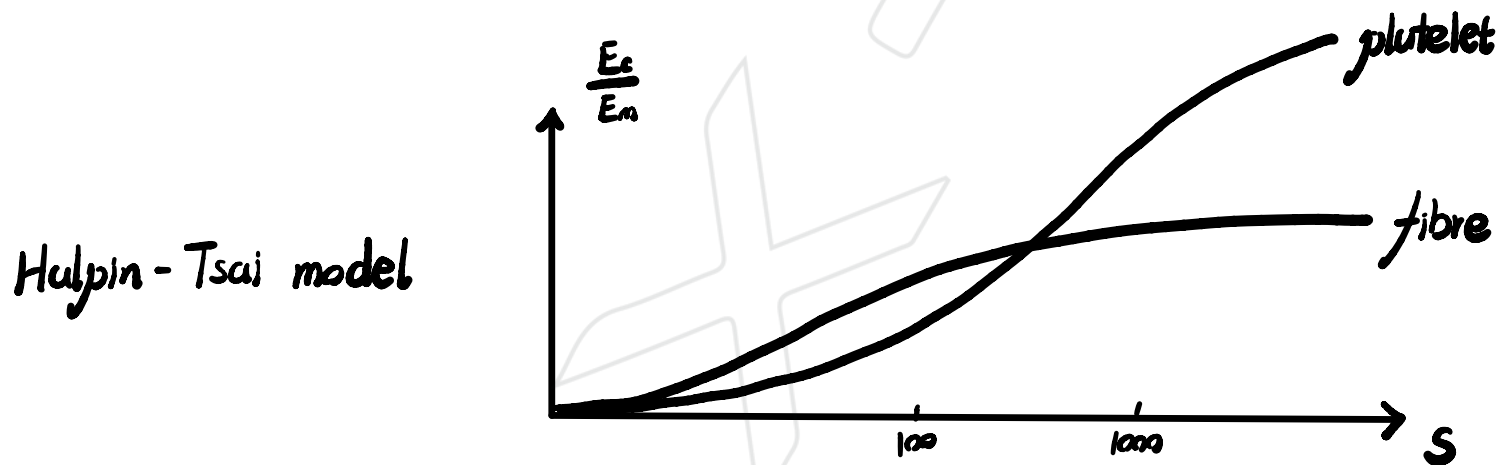
Q3

Draw an approximate plot of the composite modulus (E_x), as a function of the tilt angle ϕ , between $\phi = 0^\circ$ (x-axis along the fibre axis), $\phi = 90^\circ$ (x-axis transverse to the fibre axis).



Q4

Cite a micromechanical model that describes the variation of the Young's modulus of a composite as a function of the aspect ratio of the reinforcing fibre, and write the corresponding equation.



$$\text{where } E = \frac{E_m (1 + \xi \eta V_f)}{1 - \eta V_f} \quad \text{where } \eta = \frac{\frac{E_f}{E_m} - 1}{\frac{E_f}{E_m} + \xi}$$

Q5

Select the best composite material (fibre and matrix) and manufacturing method, for the following applications, briefly justifying your answer:

Decking for a large cruise ship

glass + epoxy

non-crimp fabrics + RIFT

Tennis racket

carbon + epoxy

bradied

**Skin for a cargo container on
civil aircraft**

aramid + epoxy

2D woven

Q6

Select the best composite material (fibre and matrix) and manufacturing method, for the following applications, briefly justifying your answer:

Hull of a yacht

carbon + epoxy

autoclave

Wind turbine blade

glass + epoxy (with wood / carbon / ... inside)

vacuum infusion

Lower drag brace for the landing gear of a fighter aircraft

carbon + epoxy

uniaxial prepeg tapes

Toilet Cover



no composite